

AUDIO QUESTION ANSWERING USING AUDIO-LANGUAGE MODEL WITH SUPERVISED FINE-TUNING AND GROUP RELATIVE POLICY OPTIMIZATION

Technical Report

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ABSTRACT

This report presents our submission for Task 5 (Audio Question Answering, AQA) of the Detection and Classification of Acoustic Scenes and Events (DCASE) 2025 Challenge. Building on the Qwen-Audio backbone, our system interprets natural language questions and generates textual answers directly from audio inputs. In our submission, an open-source variant of Qwen-Audio-7B with Group Relative Policy optimization (GRPO) is adopted as our first system (GRPO-Qwen), which is then fine-tuned on the DCASE training set with GRPO to obtain our second system (GRPO²). Experiments on the official AQA benchmark show that our systems surpass the baselines in terms of Top-1 accuracy, demonstrating their effectiveness.

Index Terms— Audio question answering, audio-language model, supervised fine-tuning, group relative policy optimization

1. INTRODUCTION

Audio Question Answering (AQA) is an audio-text cross-modal generation task that combines audio signal processing with natural language understanding [1]. Given an audio clip and a corresponding question in natural language, an AQA system must accurately locate and interpret relevant audio events, reasoning over temporal and spectral patterns to generate precise, contextually appropriate answers [2]. DCASE 2025 Challenge Task 5 focuses on AQA, providing a benchmark for evaluating the ability of the models to handle diverse acoustic scenes and queries [2, 3].

The AQA task is organized into three multi-choice subsets, each targeting a different aspect of interactive audio understanding. The **Bioacoustics QA** challenge aims to classify marine-mammal species and answer factual questions over 31 distinct vocalizations [4]. The **Temporal Soundscapes QA** challenge evaluates event detection, temporal ordering, and timestamp reasoning for 26 sound classes, drawing on the NIGENS [5], L3DAS23 [6], and TAU Spatial Sound Events 2019 [7] datasets. **Complex QA** demands higher-order inference over overlapping or abstract audio events, using clips from AudioSet [8] and Mira [9], following the MMAU benchmark [10]. Each subset includes training and development splits,

with evaluation data released for final submission, providing a comprehensive assessment of the capabilities of the models for species recognition, temporal reasoning, and abstract audio comprehension.

In this work, we develop and compare two distinct AQA systems built on the Qwen-Audio-7B backbone [3]. First, we adopt the GRPO-Qwen system [11], which is an open-source variant of Qwen-Audio-7B with GRPO on a subset of the AVQA dataset [12]. Then, we further enhance this model using GRPO on the DCASE 2025 AQA training set, yielding GRPO² AQA system. By integrating supervised learning with variance-reducing policy optimization, the GRPO² system stabilizes training and improves both generalization and answer precision.

We evaluated our systems on the official AQA benchmark provided by the DCASE Challenge. The results demonstrate that our systems achieve a significant improvement in terms of Top-1 accuracy compared to the official baselines. These results validate the effectiveness of combining a powerful audio-language backbone with a variance-reducing policy optimization scheme.

2. OUR SYSTEMS

In this section, we describe our two AQA systems submitted to DCASE 2025 Task 5, all built on the Qwen-Audio-7B backbone and progressively refined using Group Relative Policy optimization (GRPO) strategy.

2.1. GRPO-Qwen System

The GRPO-Qwen system, following the work in [11], is an open-source variant of Qwen-Audio-7B, which fine-tunes the Qwen-Audio-7B model exclusively with GRPO in a subset of 38k audio-question-answer triplets from the AVQA dataset [12]. GRPO extends the traditional reinforcement learning strategy, i.e., Proximal Policy optimization (PPO), by using each batch’s mean reward as a baseline for estimating the advantage and applying a clipped surrogate objective to limit policy shifts. This reinforcement learning-only fine-tuning skips the supervised fine-tuning (SFT) process, relying solely on reward feedback to steer model updates.

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Table 1: Performance comparison between official baseline systems and our systems.

Dataset	Accuracy (%)				
	Official Baseline			Our System	
	Qwen2-Audio-7B [3]	AudioFlamingo 2 [2]	Gemini-2.0-Flash [13]	GRPO-Qwen	GRPO ²
Dev Total	45.00	45.70	52.50	66.37	67.84
Part 1 Dev	30.00	53.90	42.00	63.84	67.86
Part 2 Dev	39.20	31.70	46.30	71.59	73.07
Part 3 Dev	49.60	49.50	56.60	64.30	65.89

2.2. GRPO² System

Furthermore, our GRPO² system applies GRPO to the GRPO-Qwen checkpoint using the DCASE 2025 Task 5 training set. We adopt the reinforcement learning configuration of our GRPO-Qwen system. This final reinforcement learning-based fine-tuning further reduces variance in policy updates and enhances both answer precision and training stability on this challenge.

3. EXPERIMENTAL RESULTS

3.1. Dataset

The DCASE 2025 Challenge Task 5 comprises three multi-choice subsets, each targeting a different facet of audio understanding. The Bioacoustics QA task contains 0.7k training and 0.2k development pairs probing species-specific vocalizations of 31 marine mammals, with sample rates from 600Hz to 160kHz and clip lengths spanning 0.4–625.0s to test fine-grained biological grounding. Temporal Soundscapes QA offers 1.0k training and 0.6k development pairs over 26 general sound classes, using 10s mono clips at 32–48 kHz to challenge models in event identification, ordering, and precise timestamp reasoning. Complex QA comprises 6.4k training and 1.6k development pairs sourced from AudioSet [8] and Mira [9], with 10s audio clips at 16kHz, requiring higher-order inference for overlapping or abstract sound events.

3.2. Performance Comparison

To further evaluate our systems, we compare them with official baselines of the challenge, i.e., Qwen2-Audio-7B [3], AudioFlamingo 2 [2] and Gemini-2.0-Flash [13], on the development dataset. As shown in Table 1, our GRPO-Qwen system significantly outperforms official baselines in terms of top-1 accuracy. It achieves an improvement of 13.87% in total accuracy, verifying the effectiveness of our GRPO-Qwen system. Meanwhile, our GRPO² system achieves an improvement of 15.34% in total accuracy.

4. REFERENCES

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